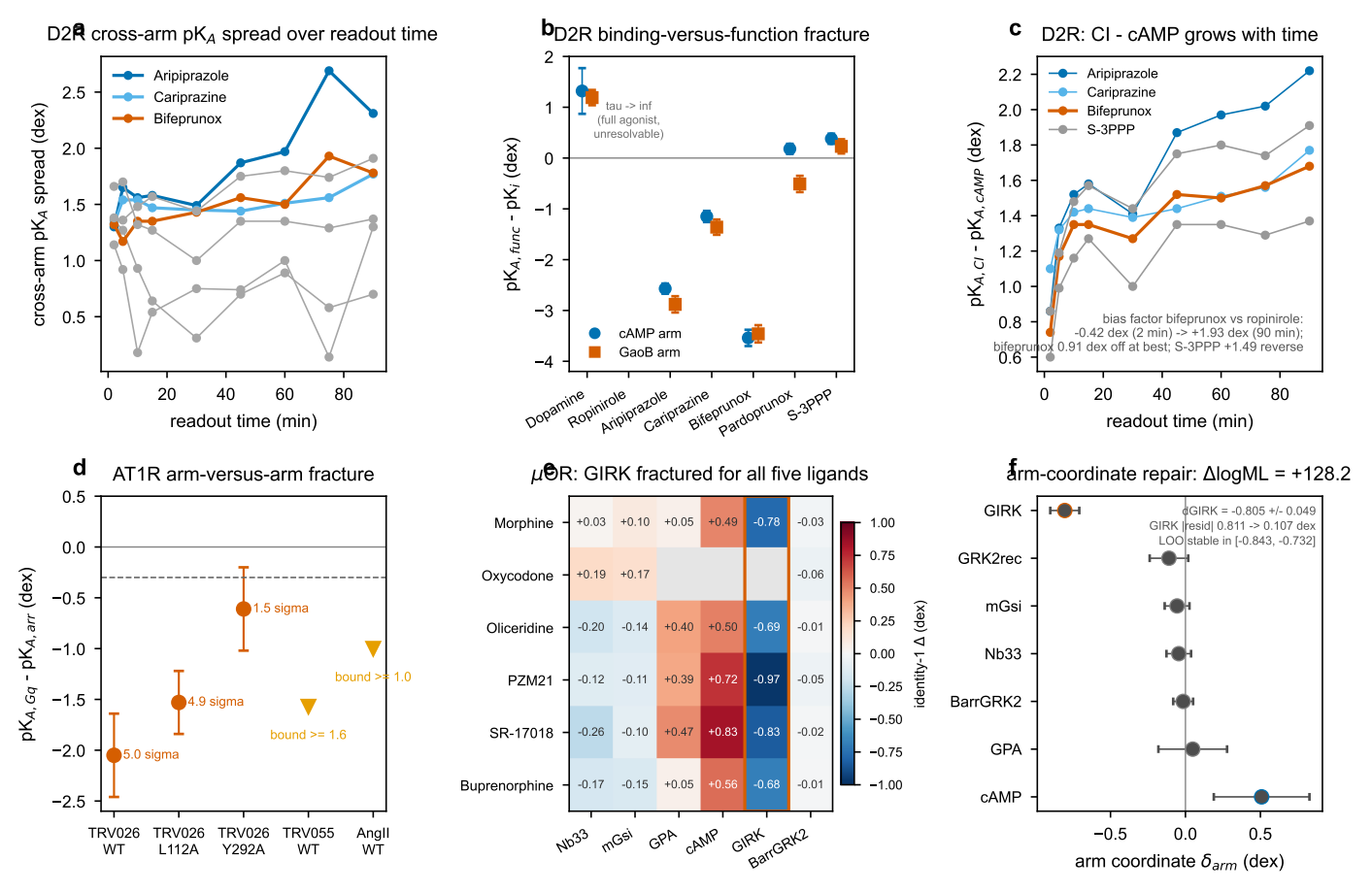

a
D2R cross-arm pK$_A$ spread over readout time
Aripiprazole
Cariprazine
Bifeprunox
cross-arm pK$_A$ spread (dex)
readout time (min)
b
D2R binding-versus-function fracture
tau -> inf
(full agonist,
unresolvable)
cAMP arm
GaoB arm
pK$_{A,func}$ - pK$_i$ (dex)
Dopamine
Ropinirole
Aripiprazole
Cariprazine
Bifeprunox
Pardoprunox
S-3PPP
c
D2R: CI - cAMP grows with time
Aripiprazole
Cariprazine
Bifeprunox
S-3PPP
bias factor bifeprunox vs ropinirole:
-0.42 dex (2 min) -> +1.93 dex (90 min);
bifeprunox 0.91 dex off at best; S-3PPP +1.49 reverse
pK$_{A,CI}$ - pK$_{A,cAMP}$ (dex)
readout time (min)
d
AT1R arm-versus-arm fracture
1.5 sigma
4.9 sigma
5.0 sigma
bound >= 1.0
bound >= 1.6
pK$_{A,Gq}$ - pK$_{A,arr}$ (dex)
TRV026 WT
TRV026 L112A
TRV026 Y292A
TRV055 WT
AngII WT
e
μOR: GIRK fractured for all five ligands
Morphine +0.03 +0.10 +0.05 +0.49 -0.78 -0.03
Oxycodone +0.19 +0.17 -0.06
Oliceridine -0.20 -0.14 +0.40 +0.50 -0.69 -0.01
PZM21 -0.12 -0.11 +0.39 +0.72 -0.97 -0.05
SR-17018 -0.26 -0.10 +0.47 +0.83 -0.83 -0.02
Buprenorphine -0.17 -0.15 +0.05 +0.56 -0.68 -0.01
Nb33
mGsi
GPA
cAMP
GIRK
BarrGRK2
identity-1 Δ (dex)
f
arm-coordinate repair: ΔlogML = +128.2
GIRK
GRK2rec
mGsi
Nb33
BarrGRK2
GPA
cAMP
dGIRK = -0.805 +/- 0.049
GIRK |resid| 0.811 -> 0.107 dex
LOO stable in [-0.843, -0.732]
arm coordinate δ_{arm} (dex)